

Yuzhou Wu

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RESEARCH INTERESTS

Actuarial Science, Insurance Economics, Optimization under Uncertainty, Risk and Decision Analysis, Quantitative Finance

EDUCATION

The University of Hong Kong Bachelor of Science in Actuarial Science	Hong Kong Sep 2023 – Jun 2027 (Expected)
- CGPA: 4.06 / 4.30. - Honors & Awards: - HKU Entrance Scholarship for Outstanding Performance (Full Award), 2023/24–2025/26 - Dean's Honors List, Faculty of Science, 2023/24–2025/26 - YF Life Scholarship in Actuarial Science and Risk Management, 2024/25–2025/26	
University of Toronto Exchange Student	Toronto, ON, Canada Sep 2026 – Dec 2026
University of California, Berkeley Summer Session	Berkeley, CA, USA Jun 2025 – Aug 2025
Coursework: Structure and Interpretation of Computer Programs (CS 61A); Data Structures and Algorithms (CS 61BL).	

RESEARCH PAPERS

- Tim J. Boonen, **Yuzhou Wu**, and Liwei Zheng. “Optimal Insurance Contract Design with Moral Hazard, Auditing, and Distortion Risk Measures.” Submitted to *Scandinavian Actuarial Journal*, 2026. Available on SSRN: doi:10.2139/ssrn.7318938. **Corresponding author.**

RESEARCH EXPERIENCE

The University of Hong Kong Research Assistant Supervisor: Prof. Tim J. Boonen	Hong Kong Jan 2025 – Aug 2026
- Developed and analyzed a two-period insurance model with endogenous policyholder effort and claim-triggered auditing under distortion risk measures including VaR and CVaR. - Derived analytical characterizations of optimal contracts and conducted numerical optimization and sensitivity analysis in MATLAB.	
Collaborative Research Project Supervisor: Prof. Fangda Liu (University of Waterloo)	Jul 2026 – Present
<i>Project: Loss-Dependent Auditing, Strategic Reporting, and Insurance Design under Moral Hazard</i> - Developing an insurance model with loss-dependent auditing and imperfect effort verification, allowing policyholders to strategically report claims under moral hazard. - Studying the joint design of audit frequency, verification quality, and insurance coverage under incentive compatibility and verification-cost constraints.	

SELECTED COURSEWORK

Mathematics: Multivariable Calculus, Linear Algebra, Discrete Mathematics, Mathematical Analysis, Real Analysis.
Probability & Statistics: Probability and Statistics, Stochastic Models, Statistical Models, Linear Models, Statistical Learning for Risk Modeling.
Actuarial Science & Quantitative Finance: Financial Mathematics, Financial Derivatives, Financial Economics, Risk Theory, Life Contingencies, Credibility Theory and Loss Modeling.
Computing & Data Science: Data Structures and Algorithms (UC Berkeley), Structure and Interpretation of Computer Programs (UC Berkeley), Computer Organization, Machine Learning, Data Science.

TECHNICAL SKILLS & QUALIFICATIONS

- Languages:** Mandarin (Native), English (IELTS: 7.5).
- Programming:** Python, R, MATLAB, C++, C, Java.
- Tools & Systems:** Git, Linux, Shell Scripting, LaTeX, TikZ, PGFPlots.
- Professional Credentials (SOA):** Passed Exam P (Probability); Passed Exam FM (Financial Mathematics); completed ASTAM requirements under the UEC pathway (pending recognition).

ADDITIONAL HONORS

- National College Entrance Examination (Gaokao): 683 / 750, Beijing, 2023.
- National Mathematical Olympiad Competition: Third Prize, 2022.